1. PI7C9X2G608GP PCIe GEN2 SWITCH — 606 MODE

NATIVE CM5 WAN1 FRONT END

2. THREE LAN7430 PCIe GIGABIT ENDPOINTS

PI7C9X2G608GP
U601A

U601A: exact PCIe lanes, reset and distributed reference clocks

WAN1: vertical VBBR-1AX1-GH PHY C7s to GND, Pad C7s NC, wired to chassis

LAN1: pin 39 tied low disables EEcat PME; pin 38 floats on its internal pull-down so advanced PCIe PM remains enabled.

LAN1: RESET_N pulled High: PCIe TX pairs AC-coupled; analog 1.2 V/13.7 volts ferrite-isolated.

LAN2: pin 39 tied low disables EEcat PME; pin 38 floats on its internal pull-down so advanced PCIe PM remains enabled.

LAN2: RESET_N pulled High: PCIe TX pairs AC-coupled; analog 1.2 V/13.7 volts ferrite-isolated.

LAN3: pin 39 tied low disables EEcat PME; pin 38 floats on its internal pull-down so advanced PCIe PM remains enabled.

LAN3: RESET_N pulled High: PCIe TX pairs AC-coupled; analog 1.2 V/13.7 volts ferrite-isolated.

Switch port 5 is disabled and unrouted to avoid PCIe Gen2 stubs and an unsafe generic test header.

LAN1: vertical VBBR-1AX1-GH PHY C7s to GND, Pad C7s NC, wired to chassis

LAN2: vertical VBBR-1AX1-GH PHY C7s to GND, Pad C7s NC, wired to chassis

LAN3: vertical VBBR-1AX1-GH PHY C7s to GND, Pad C7s NC, wired to chassis

U601B: RXN-mode strap, I2C and calibration: other straps use documented internal defaults

GPIO[5:0] = 01 selects 606 mode. GPIO1 uses its internal pulldown; GPIO0 is pulled high.

U601C: every 0.3 V / 3.3 V / ground ball shown: one 100 nF capacitor is assigned to every supply ball

HARDWARE-QUALIFIED UPSTREAM RESET

U606 asserts switch PERST# unless the CM5 request and both PCIe rail power-good signals are high.

CM5 firmware/root-complex timing must keep the request low until RST1LK and rails have been stable for at least 100 ms.

PWR_FLAG symbols declare off-sheet regulator outputs for ERC; they are not physical parts.

3. WI-FI AP MINI PCIE 4T4R

AW7915-NPI_WIFI6_4T4R
J620

AW7915-NPI: Wi-Fi 6 4T4R Mini PCIe AP; vendor maximum 9.1 W and 3.3 A recommended; dedicated rail is 3.3 V / 5 A.

J620 uses the controlled 50-07910-001 C2 land pattern; 3D/first-article fit remains a release gate.

Port 5 is disabled and unrouted; generic cable headers are prohibited on PCIe Gen2 lanes
Exact package balls/pins from Diodes DS40210 Rev 8 and Microchip DS00002631G
Native WAN1 plus three LAN7430 endpoints use exact Wurth 749911144i2 MagJacks
PI7C9X2G608GP in 606 mode: x1 upstream plus five x1 downstream ports

ProComm

Sheet: /
File: Network-PCIe.kicad_sch

Title: Radxa CM5 ProComm Carrier — PCIe / Ethernet / Wi-Fi

Size: A1	Date: 2026-08-14
KiCad E.D.A. 10.0.5-rc1	**Rev: A1**
	Id: 1/1